CHAO-JEN WONG, PH.D.

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Accomplished bioinformatician with 15+ years of experience in genomic data analysis, software development, and workflow engineering. Ph.D. in Applied Mathematics with expertise in computational genomics, epigenomics, statistical analysis, and pipeline development for NGS and long-read WGS data. Experienced in building reproducible, scalable, and portable genomic data ecosystems and integrating ML, AI, and LLM-assisted tools into bioinformatics workflows. Brings a change-ready mindset and the ability to adopt and apply new technologies to evolving scientific and computational challenges.

TECHNICAL SKILLS

Languages: R (Package Development, Bioconductor), Python, SQL, MATLAB, Bash

Workflows: Nextflow, Git, Docker/Singularity

Genomics: Fiber-seq, scRNA-seq, RNA-seq, ATAC-seq, ChIP-seq, CUT&RUN, Ribo-seq, DNA methylation, variant calling, proteomics

PROFESSIONAL EXPERIENCE

Sr. Bioinformatician — Oct 2023 – Present

Seattle Children's Research Institute

- Architected and extended the open-source Nextflow nf-core/pacvar (v1.1.0) pipeline—leveraging a globally recognized, industry-grade workflow ecosystem trusted by leading pharmaceutical companies and top academic institutions—to enable comprehensive variant calling and multi-omic methylation profiling for PacBio long-read WGS data.
- Pioneered and validated a clinical-grade PacBio long-read WGS testing to be first-tier pediatric cancer diagnostics, replacing multiple disparate molecular assays into a single, unified genomic workflow that accelerated turnaround time and significantly increased diagnostic yield.
- Deciphered the epigenetic mechanisms of a novel BMS inhibitor by engineering a CUT&RUN analysis pipeline, uncovering the specific chromatin landscape changes that govern downstream mesenchymal shift and axonogenesis.
- Integrated multi-omic and high-throughput drug screening datasets to characterize anthracycline mechanisms of action, discovering a novel chromatin-translation axis that drives anti-cancer efficacy independent of DNA damage.
- Developed software tools to strengthen multi-omics analysis and QC, including *talusR* for automated TalusBio/DIA-NN proteomics data processing and differential analysis with limma, and *peakable* for CUT&RUN replicate QC using cosine similarity and embedding-matrix-based PCA to reduce misleading downstream conclusions.

Bioinformatician III — Oct 2015 – Jun 2023

Tapscott Lab, Human Biology Department, Fred Hutchinson Cancer Center

- Led data-driven research for a cross-functional initiative that discovered the primary disease progression mechanisms in facioscapulohumeral muscular dystrophy (FSHD), published in Human Molecular Genetics.
- Identified the first validated panels of gene signatures representing FSHD molecular activity, setting the industry standard for post-therapeutic clinical trial measurements.
- Pioneered research demonstrating immune cell infiltration in FSHD-affected muscle, shifting the scientific understanding of the disease's systemic progression.
- Founded and led a peer-to-peer data science community at Fred Hutch, growing membership from 32 to over 250 scientists within a year.

Bioinformatics Analyst II — Oct 2011 – Oct 2015

Grady Lab, Clinical Research Division, Fred Hutch Cancer Center

- Contributed to the first published study reporting human colon DNA methylation profiles and subtype classification in colon adenocarcinoma, using Illumina 450K array-based methylation profiling.
- Collaborated on two peer-reviewed studies on DNA methylation-based modeling of the human biological clock and epidemiologic analysis of cigarette smoking associations with prostate cancer outcomes.

Bioconductor Core Developer **Sep 2008 – Sep 2011**

Gentleman Lab, Computational Biology Program, Fred Hutchinson Cancer Center

- Contributed to the core infrastructure of Bioconductor for genomic data analysis and visualization.
- Developed novel algorithms for multi-channel flow cytometry normalization using functional data analysis.
- Served as a lead reviewer for 50+ Bioconductor package submissions, enforcing rigorous standards for statistical validity and software quality.

Postdoctoral Researcher **Aug 2007 – May 2008**

San Diego State University

- Research fields included numerical methods for solving partial differential equations and microarray analysis investigating associations between antidiabetic medications and cardiovascular disease.

Analytic Consultant **Apr 2006 – Jan 2007**

Pacific Nanotechnology (Contract)

- Developed atomic force microscopy tip deconvolution and image reconstruction algorithms to correct imaging artifacts caused by imperfect probe-tip geometry, resulting in a peer-reviewed publication.

Academic Part-Time Engineer **Apr 2000 – Sep 2005**

Jet Propulsion Laboratory

- Developed high-fidelity Telecom Forecaster Predictor (TFP) software in MATLAB to simulate communication links for Mars Rover and Deep Impact missions.
- Co-authored volumes in JPL's Deep Space Communications and Navigation Center of Excellence (DESCANSO) book series.

EDUCATION

Ph.D. & M.S. in Mathematics **2001 – 2006**

Claremont Graduate University

Relevant coursework: statistical inference, time series analysis, and partial differential equations.
Thesis: Simulation of Immobilized Enzyme Kinetics and Transport in Sessile Hydrogen Drops. Adviser: Ali Nadim.

M.A. & B.A. in Applied Mathematics **1996 – 2000**

California State University, Fullerton

RECENT PUBLICATIONS

• Sadeeshkumar H, Wong CJ, et al. Anthracyclines target the chromatin-translation axis independent of DNA damage. Manuscript under review at *Nature*.

• Wong CJ, et al. Regional and bilateral MRI and gene signatures in facioscapulohumeral dystrophy. *Human Molecular Genetics*. 2024;33(8):698-708.

• Hamm DC, Paatela EM, Bennett SR, Wong CJ, et al. The transcription factor DUX4 orchestrates translational reprogramming. *PLOS Biology*. 2023;21(9):e3002317.

• Wong CJ, et al. Canine DUXC: implications for DUX4 retrotransposition and preclinical models of FSHD. *Human Molecular Genetics*. 2022;31(10):1694-1704.

PROFESSIONAL ACTIVITIES

• Nextflow Ambassador, 2026.

• Poster presenter, Nextflow Summit Boston, 2026.

• Contributor to nf-core/modules and participant in Nextflow Hackathons, 2024-2026.

COURSERA CERTIFICATIONS

• Sequence Models (DeepLearning.AI)

• Convolutional Neural Networks (DeepLearning.AI)

• Neural Networks and Deep Learning (DeepLearning.AI)

• Applied Machine Learning in Python (U. Michigan)

- SQL Basics for Data Science Specialization (UC Davis)